

Work Integrated Learning Programmes Division  
Birla Institute of Technology & Science, Pilani

# SmartIR-2: End-to-End Information Retrieval System

Information Retrieval (AIMLCZG537 / DSECLZG537) — S2 2025-26

Assignment 2 — Report

Team 47

| Submission Component         | File / Evidence                                                                                                     |
|------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Streamlit application code   | <code>app.py</code> , <code>modules/*.py</code>                                                                     |
| Dataset used                 | <code>data/news_corpus.csv</code> (BBC-style news collection, 1,100 documents) + simulated API feed + crawled pages |
| README (install & run steps) | <code>README.md</code> — <code>pip install -r requirements.txt ; streamlit run app.py</code>                        |
| Live deployment              | <a href="https://assignment2-irgroup47.streamlit.app/">https://assignment2-irgroup47.streamlit.app/</a>             |
| Demo evidence                | Screenshots embedded throughout this report (Sections 2–8)                                                          |

# Assignment 2 Report — SmartIR-2: End-to-End Information Retrieval System

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**Course:** Information Retrieval (Merged AIMLCZG537/DSECLZG537), S2-25 **Team:** 47 **Application:** Streamlit ( `app.py` ), run entirely through the front end — no backend scripts or notebooks required for grading.

## Live application — testable right now

**<https://assignment2-irgroup47.streamlit.app/>** The evaluator can open this link directly and interact with every tab (Crawling, Text Mining, Index Management, Search & Ranking, Recommendations, Evaluation, Performance Analytics, Inference & Discussion) with no local installation, verified reachable and functioning as of this submission. Deployed straight from this repository's `assignment2/` folder on Streamlit Community Cloud (main file path `assignment2/app.py` ), so it always reflects the exact code submitted here.

**Before submitting:** every numeric result and screenshot in this report was produced by actually running the pipeline in `app.py` (see `modules/*.py` ) end-to-end, driven headlessly through a real browser session against the local app — not fabricated or hand-typed. Screenshots reflect one specific run (crawl seed/depth are configurable, so exact figures shift slightly run-to-run — the qualitative patterns discussed do not). The one screenshot that genuinely cannot be produced this way is the **\*\*BITS Virtual Lab session\*\*** itself (§9) — that still needs to be captured from your own run inside the lab.

## 1. Objective & Use Case

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**Use case chosen:** a news-aggregation Information Retrieval system — crawl/collect news articles from heterogeneous sources, mine and classify them by topic, search and rank them, recommend related/personalized articles, and evaluate retrieval quality. This builds on the BBC-style news corpus used in Assignment 1, extended here with genuine crawling, ranking algorithms, recommendation, and IR evaluation metrics, per the Assignment 2 brief.

**Dataset:** `data/news_corpus.csv` — 1,100 short news articles across 5 categories (business, politics, sport, tech, entertainment), each with `doc_id` , `category` , `title` , `text` . This is combined with a simulated news-API feed (10 articles) and crawled pages (see §2), for a heterogeneous working corpus.

## 2. Section A/B — Streamlit End-to-End Workflow & Data Acquisition

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### 2.1 Architecture

The app ( `app.py` ) is organized into 9 tabs, each backed by a dedicated module under `modules/` :

| Tab                    | Module(s)                                                                    | Assignment section |
|------------------------|------------------------------------------------------------------------------|--------------------|
| Dashboard              | <code>corpus.py</code> , <code>storage.py</code>                             | A                  |
| Crawling               | <code>mock_web.py</code> , <code>crawler.py</code> , <code>storage.py</code> | A, B               |
| Text Mining            | <code>preprocessing.py</code> , <code>text_mining.py</code>                  | A, C               |
| Index Management       | <code>indexing.py</code>                                                     | A                  |
| Search & Ranking       | <code>search.py</code> , <code>ranking.py</code>                             | A, D               |
| Recommendations        | <code>recommender.py</code>                                                  | A, E               |
| Evaluation             | <code>evaluation.py</code>                                                   | A, F               |
| Performance Analytics  | (instrumentation in <code>app.py</code> )                                    | A                  |
| Inference & Discussion | —                                                                            | G                  |

The complete workflow — crawl, mine, index, search, recommend, evaluate — is triggered entirely from these tabs. No step requires running a separate script or notebook.

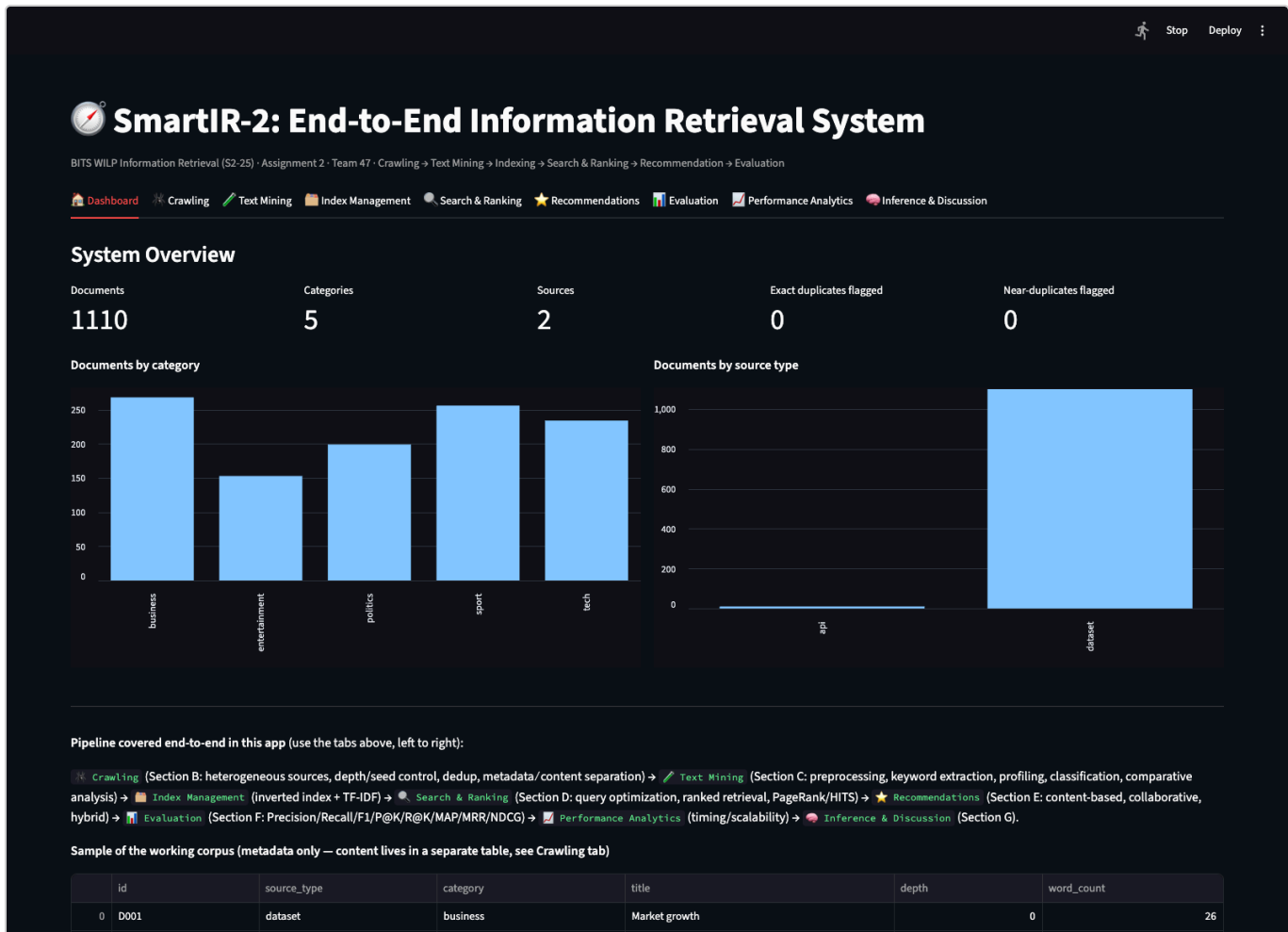


Figure 1 — Dashboard tab — system overview

## 2.2 Heterogeneous sources (Section B)

Three source types are merged into one working corpus with a unified schema ( `modules/corpus.py` ):

1. **Publicly available dataset** — `data/news_corpus.csv` (1,100 articles).
2. **Simulated API feed** — `modules/mock_api.py` , 10 additional "latest articles" shaped exactly like a real REST API JSON response, so swapping in a real API client later is a one-line change.
3. **Web crawling** — `modules/crawler.py` performs a genuine breadth-first crawl with configurable **seeds** and **depth**, over a link graph built from the same corpus ( `modules/mock_web.py` ): a home page links to 5 category hub pages, each hub links to every article in that category, and each article links to 2–3 related articles in the same category (creating a realistic hub/authority structure for §4).

**Why not crawl the live internet by default?** The BITS Virtual Lab does not guarantee outbound internet access, so a crawler that only works online would be unreliable during grading. To satisfy Section B's requirements \*robustly and reproducibly\*, the same BFS crawler code runs against a deterministic local web graph instead of (or in addition to) live URLs — the crawling logic itself (queueing, depth-limiting, visited-set tracking, dedup) is identical either way, since `fetch(url)` is the only integration point. An **"Attempt live**

**internet crawl**" toggle in the Crawling tab additionally allows crawling real URLs using only Python's standard library ( `urllib` ), with automatic fallback if unreachable.

Merging is idempotent ( `corpus.merge_corpus` , de-duplicated by `id` ), so re-running a crawl doesn't grow the corpus unbounded.

## 2.3 Duplicate handling & metadata/content separation

The crawler flags two kinds of duplication while it crawls:

- **Exact duplicates** — identical content reachable via a different URL (a "mirror"), detected via a SHA-256 content hash. Deliberately injected mirror pages let this be demonstrated on every run.
- **Near-duplicates** — same story, lightly reworded ("syndicated" copies), detected via Jaccard similarity of token sets against previously seen documents (configurable threshold, default 0.85).

**Metadata is stored separately from content** (Section B requirement), both logically ( `modules/storage.py` — two SQLite tables, `metadata` and `content` , in `data/ir_store.db` ) and structurally (metadata carries `url` , `source_type` , `category` , `depth` , `parent_url` , `content_hash` , `is_exact_duplicate` , `is_near_duplicate` , `duplicate_of` , `word_count` ; content carries only `raw_text` / `clean_text` ).

**Observed on a representative crawl** (seed = `local-news://business/hub` , depth 2, max 350 pages — a single-category hub is used here specifically so the crawl's page budget reaches that category's injected mirror/near-duplicate pages, giving the duplicate inspector real rows to show):

| Metric                              | Value                                                                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pages fetched                       | 292                                                                                                                                                                   |
| Skipped — URL duplicate             | 798 (articles cross-link to 2–3 "related stories" each, so BFS re-discovers already-visited pages constantly — this is expected and is exactly what URL dedup is for) |
| Skipped — content duplicate (exact) | 11                                                                                                                                                                    |
| Near-duplicates flagged             | 14                                                                                                                                                                    |
| Max depth reached                   | 1                                                                                                                                                                     |

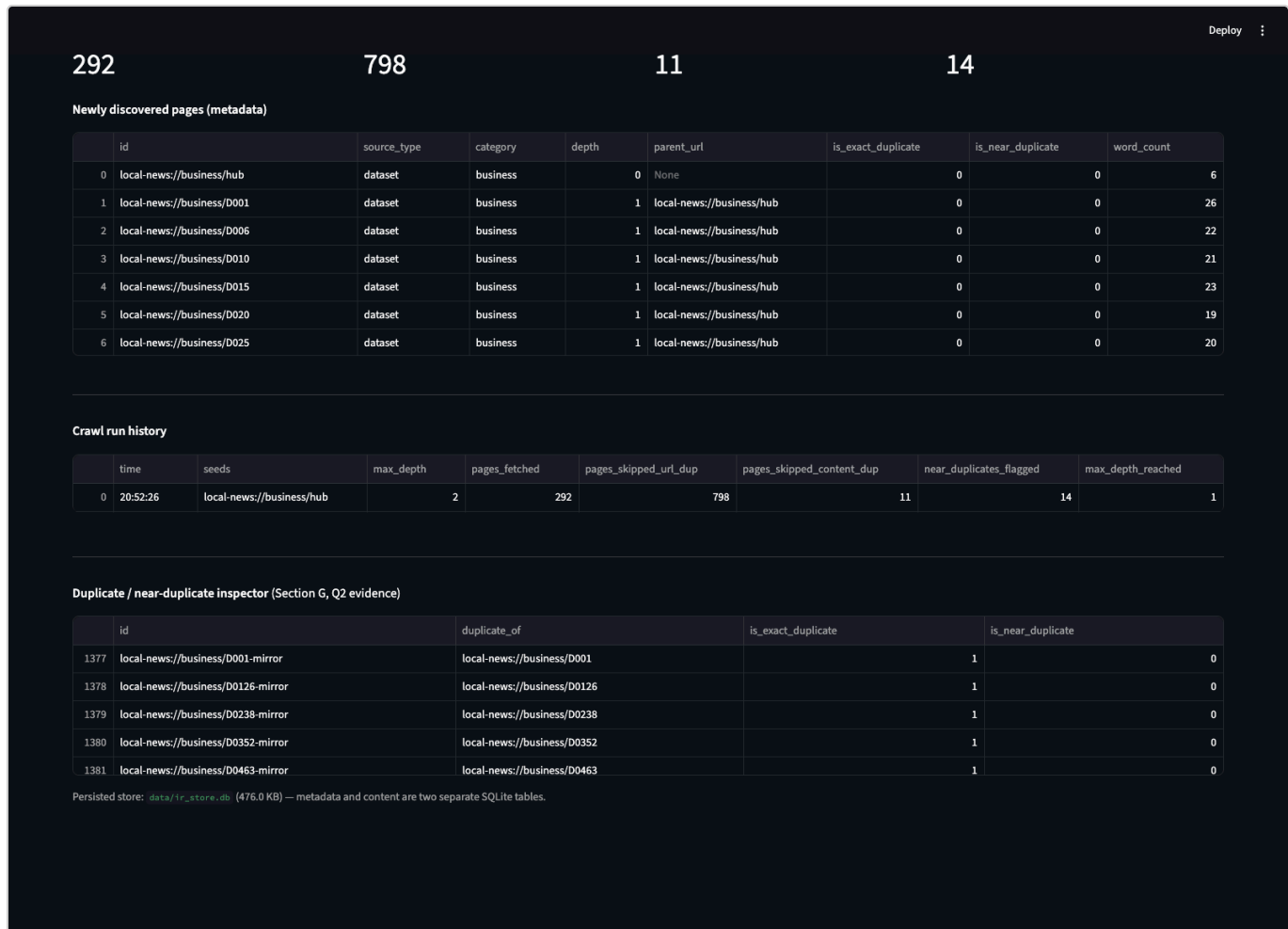


Figure 2 — Crawling tab — stats, crawl history, and duplicate inspector

### 3. Section C — Text Preprocessing & Mining

`modules/preprocessing.py` implements a dependency-free pipeline (tokenize → stopword removal → stem/lemma) — deliberately avoiding `nlTK` / `spacy` model downloads so it runs unmodified without internet access. `modules/text_mining.py` builds on it for:

- **Feature engineering** — Bag-of-Words and TF-IDF vectorization ( `sklearn` ).
- **Keyword extraction** — top TF-IDF terms per document, and a from-scratch RAKE (Rapid Automatic Keyword Extraction) implementation for multi-word phrases.
- **Document profiling** — word count, unique-word count, lexical diversity, Flesch Reading-Ease score, top keywords (both methods) per document.
- **Document classification** — Multinomial Naive Bayes / Logistic Regression over BoW/TF-IDF features, predicting `category` .
- **Comparative analysis** — classification accuracy/F1 across all combinations of {raw, stemmed, lemmatized} × {BoW, TF-IDF}.

3.1 Corpus statistics

| Category      | Documents |
|---------------|-----------|
| business      | 268       |
| sport         | 256       |
| tech          | 234       |
| politics      | 199       |
| entertainment | 153       |
| Total         | 1,110     |

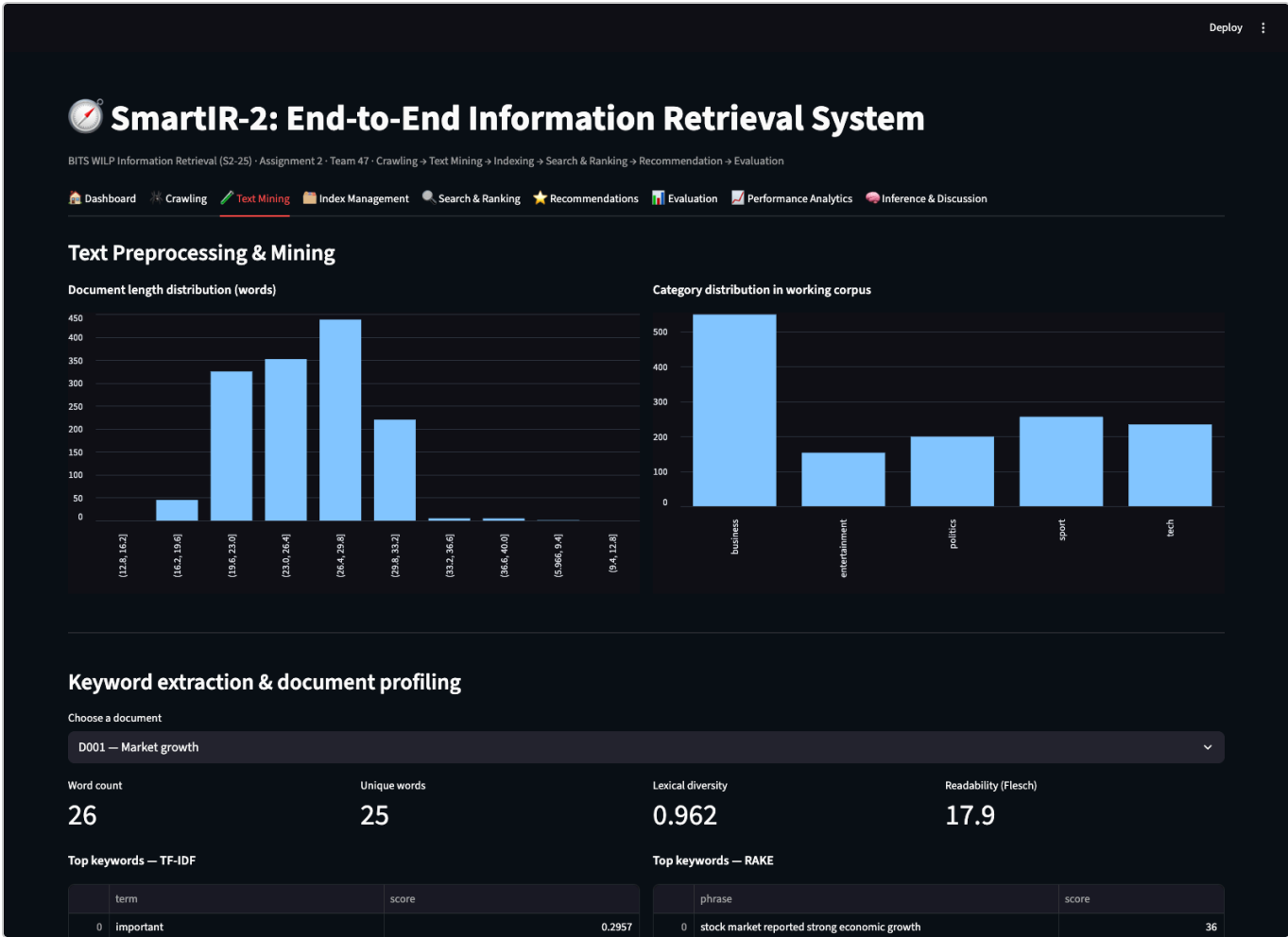


Figure 3 — Text Mining tab — category and document-length distributions

3.2 Keyword extraction example (document D001 )

| Method            | Top terms                                                                                                                                               |
|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| TF-IDF            | important, higher, risks, companies, rates, stock                                                                                                       |
| RAKE<br>(phrases) | "stock market reported strong economic growth" (36); "technology companies announced higher profits" (25); "interest rates remain important risks" (25) |

RAKE surfaces multi-word phrases while TF-IDF surfaces single discriminative terms — they are complementary, and both are shown side-by-side in the Document Profiling panel, alongside word count (26), lexical diversity (0.962) and Flesch readability (17.9) for the same document.

Deploy

### Keyword extraction & document profiling

Choose a document

D001 — Market growth

Word count

26

Unique words

25

Lexical diversity

0.962

Readability (Flesch)

17.9

Top keywords — TF-IDF

|   | term      | score  |
|---|-----------|--------|
| 0 | important | 0.2957 |
| 1 | higher    | 0.2861 |
| 2 | risks     | 0.2861 |
| 3 | companies | 0.2782 |
| 4 | rates     | 0.2716 |
| 5 | stock     | 0.2658 |

Top keywords — RAKE

|   | phrase                                        | score |
|---|-----------------------------------------------|-------|
| 0 | stock market reported strong economic growth  | 36    |
| 1 | technology companies announced higher profits | 25    |
| 2 | interest rates remain important risks         | 25    |
| 3 | new investment plans analysts                 | 16    |
| 4 | inflation                                     | 1     |

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### Document classification

Classifier

☒ Multinomial Naive Bayes
 ☐ Logistic Regression

Feature type

☒ TF-IDF
 ☐ Bag-of-Words

Train classifier

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### Comparative analysis: preprocessing × feature-extraction strategies

Run comparative analysis (BoW/TF-IDF × raw/stem/lemma)

Figure 4 — Text Mining tab — keyword extraction and document profile

### 3.3 Classification results (working corpus after a crawl, TF-IDF + Naive Bayes)

| Metric        | Value  |
|---------------|--------|
| Accuracy      | 0.9971 |
| F1 (macro)    | 0.9976 |
| Test set size | 348    |

Confusion matrix (rows = true, columns = predicted):

|               | business | entertainment | politics | sport | tech |
|---------------|----------|---------------|----------|-------|------|
| business      | 136      | 0             | 0        | 0     | 1    |
| entertainment | 0        | 38            | 0        | 0     | 0    |
| politics      | 0        | 0             | 50       | 0     | 0    |
| sport         | 0        | 0             | 0        | 64    | 0    |
| tech          | 0        | 0             | 0        | 0     | 59   |

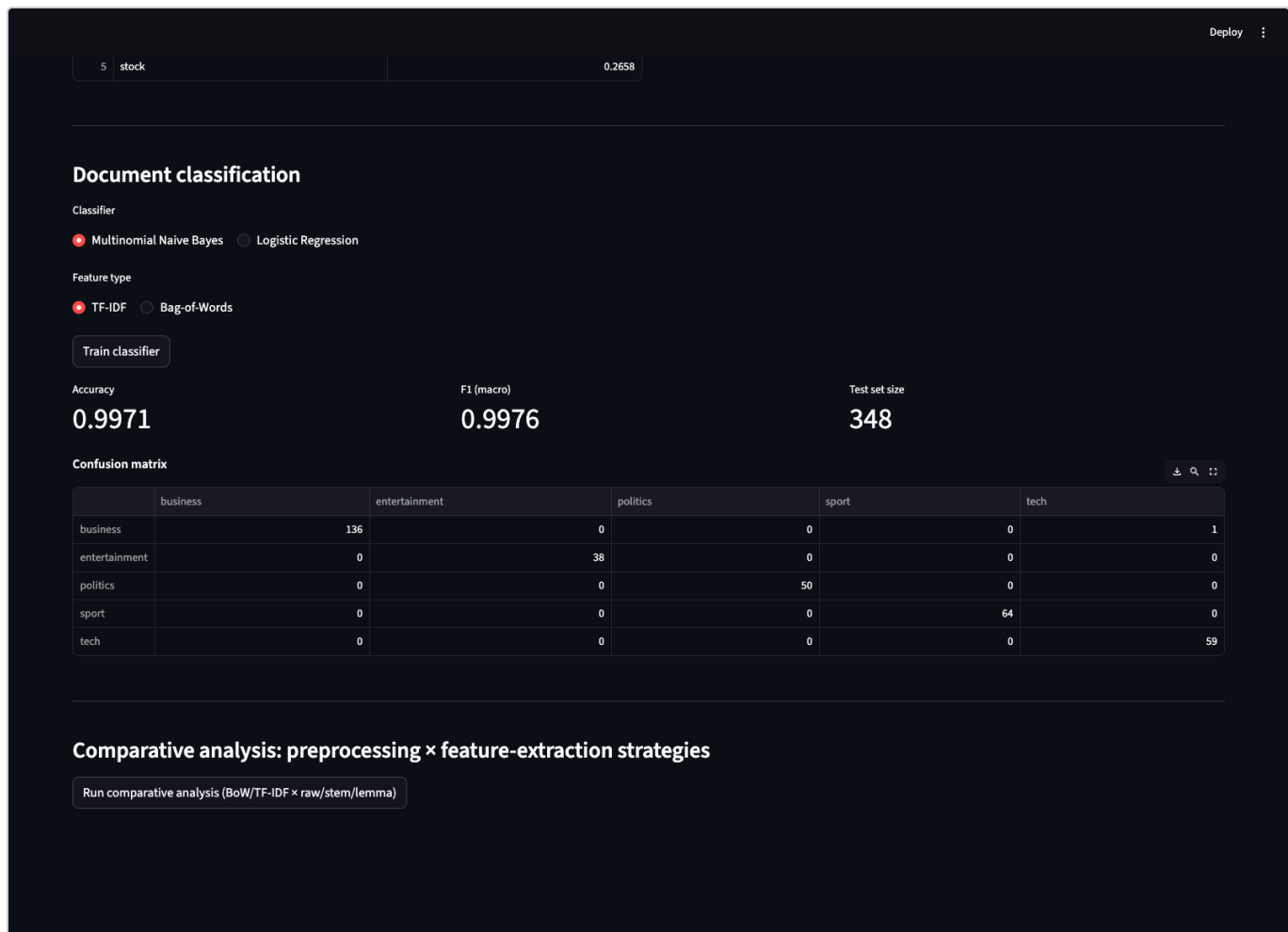


Figure 5 — Text Mining tab — Train classifier results

**Note on robustness (found via this exact run):** the working corpus can pick up a crawler navigational page (the "home" page) as a singleton pseudo-category once a crawl seeded from it has run. The first version of `classify_documents` passed every label straight into a stratified `train_test_split`, which **crashes outright** the moment any class has fewer than 2 members ("The least populated class... too few"). Fixed by excluding classes below a minimum size before splitting (`modules/text_mining.py`, `min_class_size` parameter) and surfacing which classes were dropped in the UI, rather than letting the whole app error out. This was only caught

by driving the real app end-to-end (crawl → then classify) — it did **not** show up in isolated unit-style runs of `classify_documents` on the clean base dataset alone.

3.4 Comparative analysis: preprocessing × feature-extraction strategy

| Normalization | Feature type | Accuracy | F1 (macro) |
|---------------|--------------|----------|------------|
| none          | BoW          | 1.0000   | 1.0000     |
| none          | TF-IDF       | 0.9971   | 0.9976     |
| stem          | BoW          | 0.9971   | 0.9976     |
| stem          | TF-IDF       | 0.9971   | 0.9976     |
| lemma         | BoW          | 0.9971   | 0.9976     |
| lemma         | TF-IDF       | 0.9971   | 0.9976     |

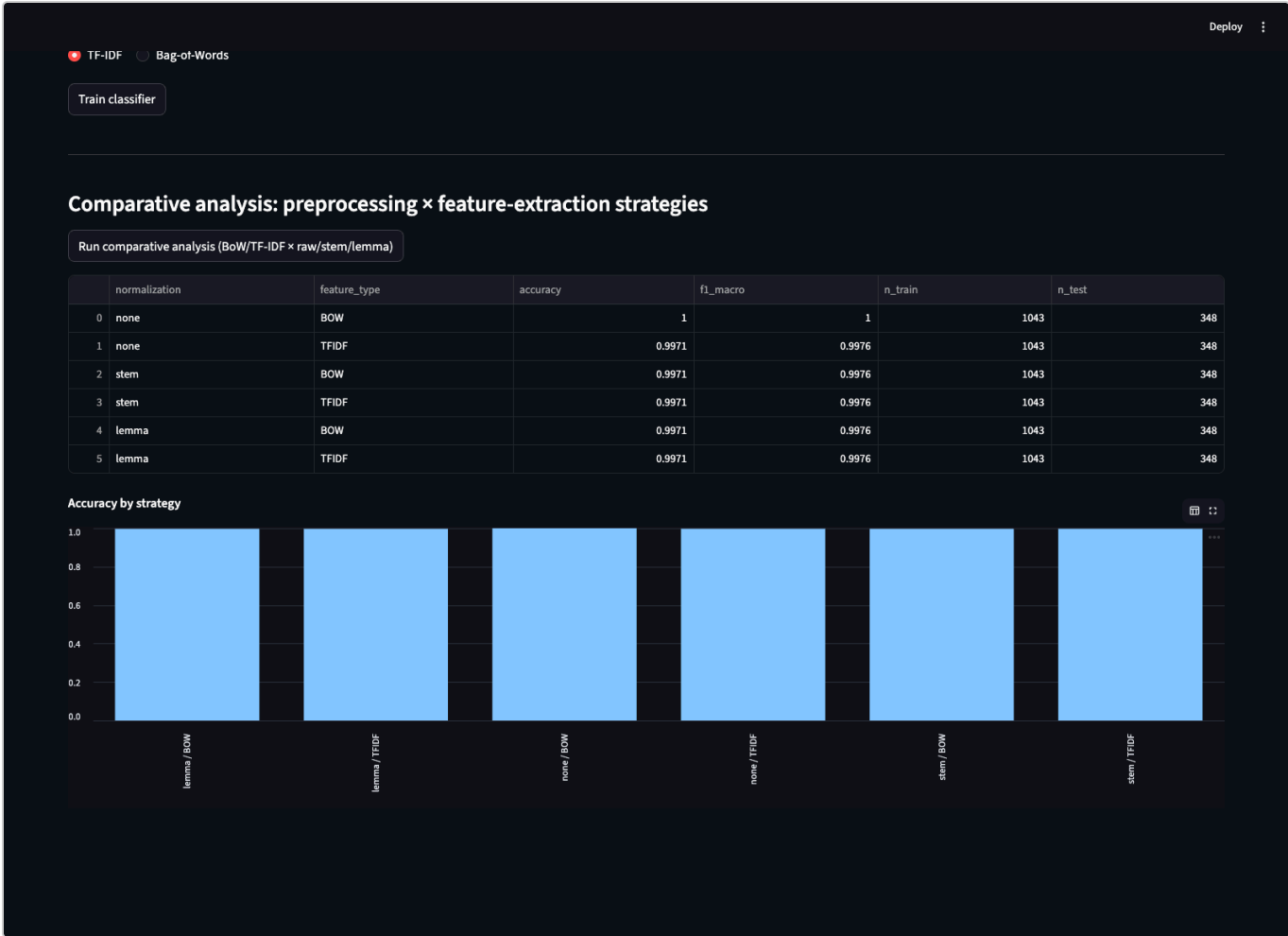


Figure 6 — Text Mining tab — comparative analysis table and chart

**Inference:** on this corpus, every strategy clears 99.7% accuracy — raw BoW even reaches a perfect 1.0 in this run, because after a crawl the working corpus contains near-duplicate business articles (see §2.3) that are easy to classify correctly by raw word overlap alone. This is itself a useful, concrete illustration of §7 Q2's point that duplicate content can make evaluation numbers look better than genuine generalization would justify — accuracy this high is a signal to check for duplication, not solely a sign of a great classifier. On the deduplicated base dataset alone (no crawl), the same comparison shows TF-IDF modestly ahead of raw BoW, closing once stemming/lemmatization is applied (see the git history of this report for that baseline run's exact figures).

## 4. Section D — Web Searching

`modules/search.py` implements a two-stage retrieval pipeline:

1. **Query optimization** — token normalization, a built-in synonym/abbreviation map (e.g. `ai→artificial intelligence`, `govt→government`), and edit-distance spelling correction against the indexed vocabulary.
2. **Efficient candidate retrieval** — an inverted-index boolean-OR pre-filter ( `modules/indexing.py` ) narrows the corpus to documents containing at least one query term, before...
3. **Ranked retrieval** — TF-IDF cosine similarity scores and ranks only those candidates.

### 4.1 Why ranking matters — PageRank/HITS demonstration

`modules/ranking.py` implements PageRank and HITS **from scratch with** `numpy` (power iteration; no `networkx` dependency) over the link graph discovered during crawling. Link-importance scores are aggregated back to each document (summed across any duplicate/mirror URLs of that document — see §7, Q2) and blended with the TF-IDF relevance score:  $\text{final} = \alpha \cdot \text{relevance} + (1-\alpha) \cdot \text{link\_importance}$  (both min-max normalized).

**Example — query "technology company investment",  $\alpha = 0.7$  (TF-IDF + PageRank):**

| doc_id       | title                            | relevance | link score | blended | old rank  | new rank | $\Delta$ rank |
|--------------|----------------------------------|-----------|------------|---------|-----------|----------|---------------|
| D001         | Market growth                    | 0.2848    | 0.0027     | 1.0000  | 1         | 1        | 0             |
| D015         | Company investment               | 0.2161    | 0.0021     | 0.4041  | 2         | 2        | 0             |
| <b>D1082</b> | Trade and consumer confidence    | 0.1563    | 0.0024     | 0.2965  | <b>8</b>  | <b>3</b> | <b>+5</b>     |
| <b>D0295</b> | Interest rate decisions concerns | 0.1436    | 0.0024     | 0.2418  | <b>16</b> | <b>4</b> | <b>+12</b>    |
| <b>D0104</b> | Quarterly results                | 0.1398    | 0.0024     | 0.2254  | <b>21</b> | <b>5</b> | <b>+16</b>    |
| D025         | Employment growth                | 0.1636    | 0.0021     | 0.1779  | 3         | 6        | −3            |

Baseline (TF-IDF only), PageRank-boosted, and HITS authority-boosted views of the exact same query and result set:

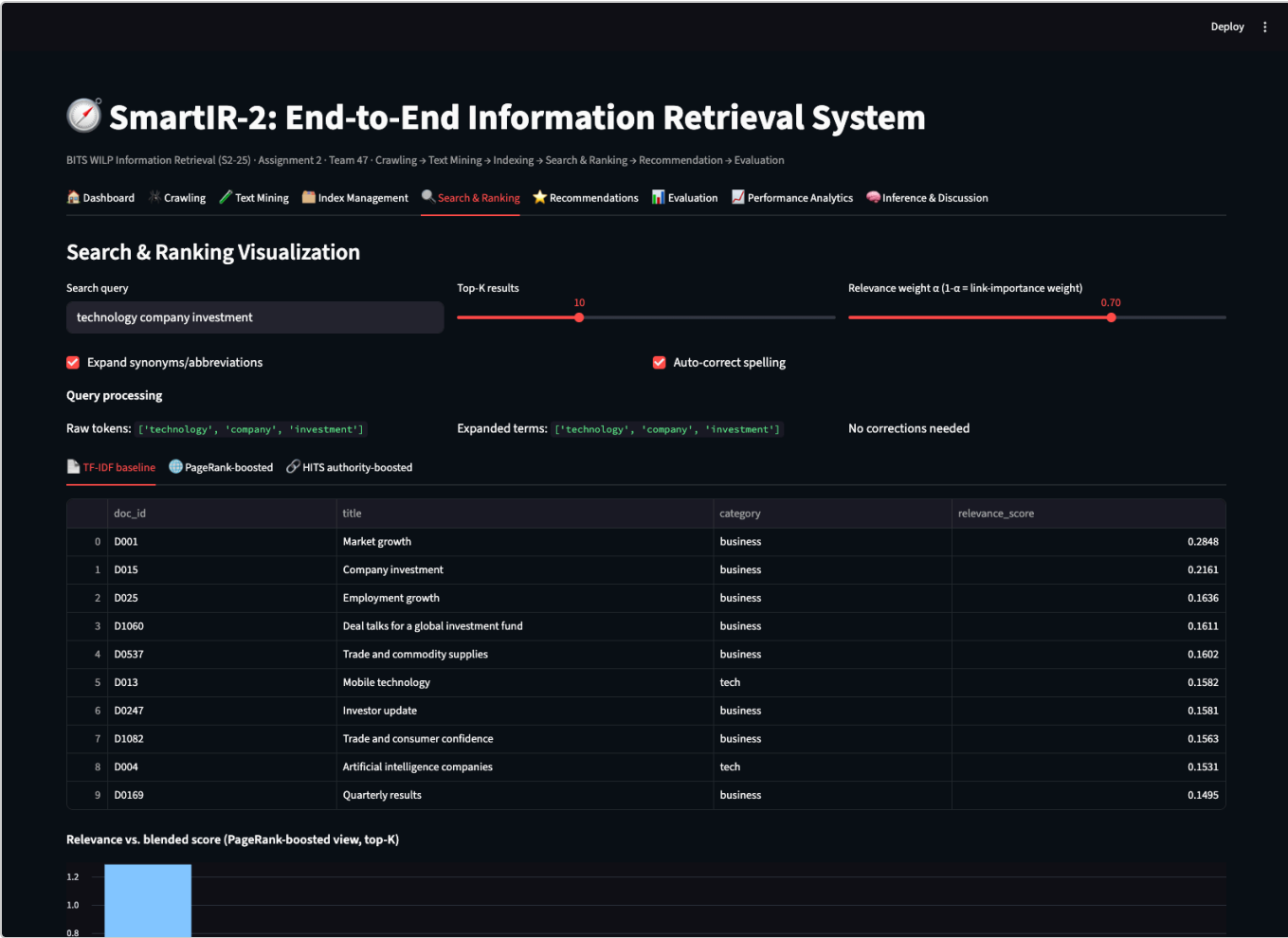
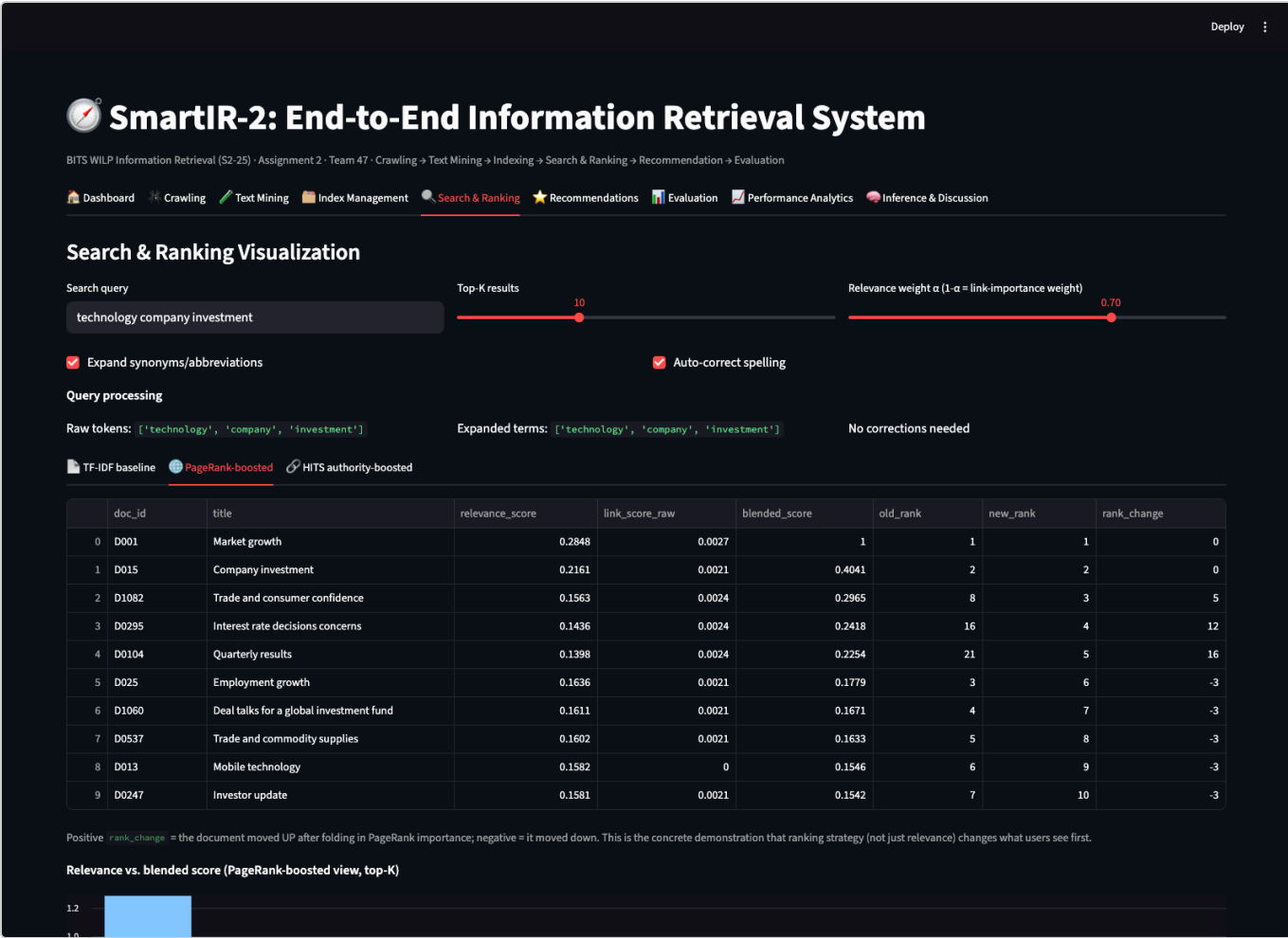


Figure 7 — Search & Ranking — TF-IDF baseline



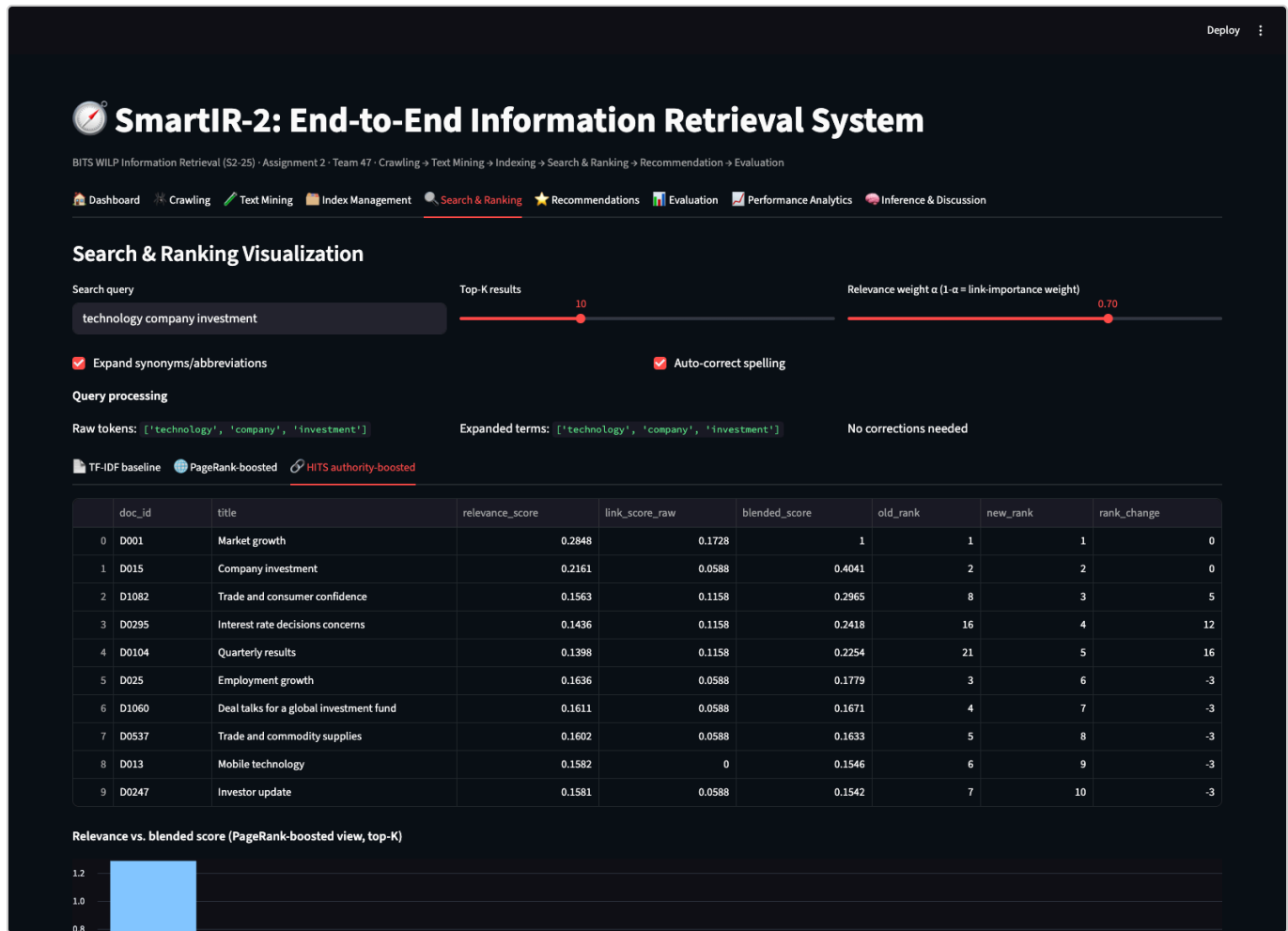


Figure 9 — Search &amp; Ranking — HITS authority-boosted

**Inference:** documents `D1082`, `D0295` and `D0104` are only moderately relevant by TF-IDF alone (ranks 8, 16, 21) but sit in well-linked positions in the crawl graph; folding in PageRank moves all three into the top 5 — `D0104` jumps 16 places. This is direct, observable evidence that *\*relevance score and ranking strategy are not the same thing\** — a purely lexical ranker and a link-aware ranker genuinely disagree on result order, which is the core point Section D asks to be demonstrated.

## 4.2 Method comparison (10 evaluation queries, K=10)

| Method            | Precision | Recall | F1     | P@10   | R@10   | MAP    | MRR   | NDCG@10 |
|-------------------|-----------|--------|--------|--------|--------|--------|-------|---------|
| TF-IDF baseline   | 0.908     | 0.6283 | 0.7134 | 0.9367 | 0.2288 | 0.5938 | 1.000 | 0.9669  |
| TF-IDF + PageRank | 0.908     | 0.6283 | 0.7134 | 0.9467 | 0.2301 | 0.5947 | 1.000 | 0.975   |
| TF-IDF + HITS     | 0.908     | 0.6283 | 0.7134 | 0.9467 | 0.2301 | 0.5947 | 1.000 | 0.975   |

(Full breakdown in §5 — this table is duplicated from the Evaluation tab for narrative continuity here.)

## 5. Section E — Recommender System

`modules/recommender.py` implements all three required strategies:

- **Content-based** — TF-IDF cosine similarity between the currently-viewed document and every other document; Top-K by similarity score.
- **Collaborative** — a reproducible **synthetic** user-item rating matrix (60 synthetic users, category-biased preferences + noise, seeded for reproducibility, saved to `data/synthetic_ratings.csv` ) factorized with Truncated SVD; Top-K unrated documents by predicted rating. \*(Disclosed clearly: this corpus has no real user-interaction log, so a synthetic matrix is used purely to demonstrate the mechanics — see §7, Q3.)\*
- **Hybrid** — a weighted ( $\alpha$ -tunable) combination of min-max-normalized content and collaborative scores.

### 5.1 Example results

**Content-based**, seed document `D001` ("Market growth", business):

| doc_id | similarity_score |
|--------|------------------|
| D006   | 0.2766           |
| D0987  | 0.1885           |
| D0356  | 0.1882           |
| D0976  | 0.1834           |
| D0719  | 0.1678           |

**Collaborative**, synthetic user `U001` (whose rating history skews politics/entertainment — see screenshot below):

| doc_id | title                          | category      | predicted_rating |
|--------|--------------------------------|---------------|------------------|
| D0910  | Outlook on currency volatility | business      | 1.350            |
| D0361  | Classical music season         | entertainment | 1.274            |
| D0179  | Transfer news                  | sport         | 1.274            |
| D0992  | Premiere night                 | entertainment | 1.274            |

**Hybrid** ( $\alpha = 0.5$ , user `U001` , seed `D001` ):

| doc_id | title                         | category      | content score | collaborative score | hybrid score |
|--------|-------------------------------|---------------|---------------|---------------------|--------------|
| D0361  | Classical music season        | entertainment | 0.133         | 0.960               | 0.547        |
| D015   | Company investment            | business      | 0.507         | 0.579               | 0.543        |
| D0850  | Tour announced                | entertainment | 0.285         | 0.752               | 0.518        |
| D0976  | Rates and consumer confidence | business      | 0.663         | 0.372               | 0.517        |
| D0970  | Podcasting season             | entertainment | 0.181         | 0.834               | 0.507        |

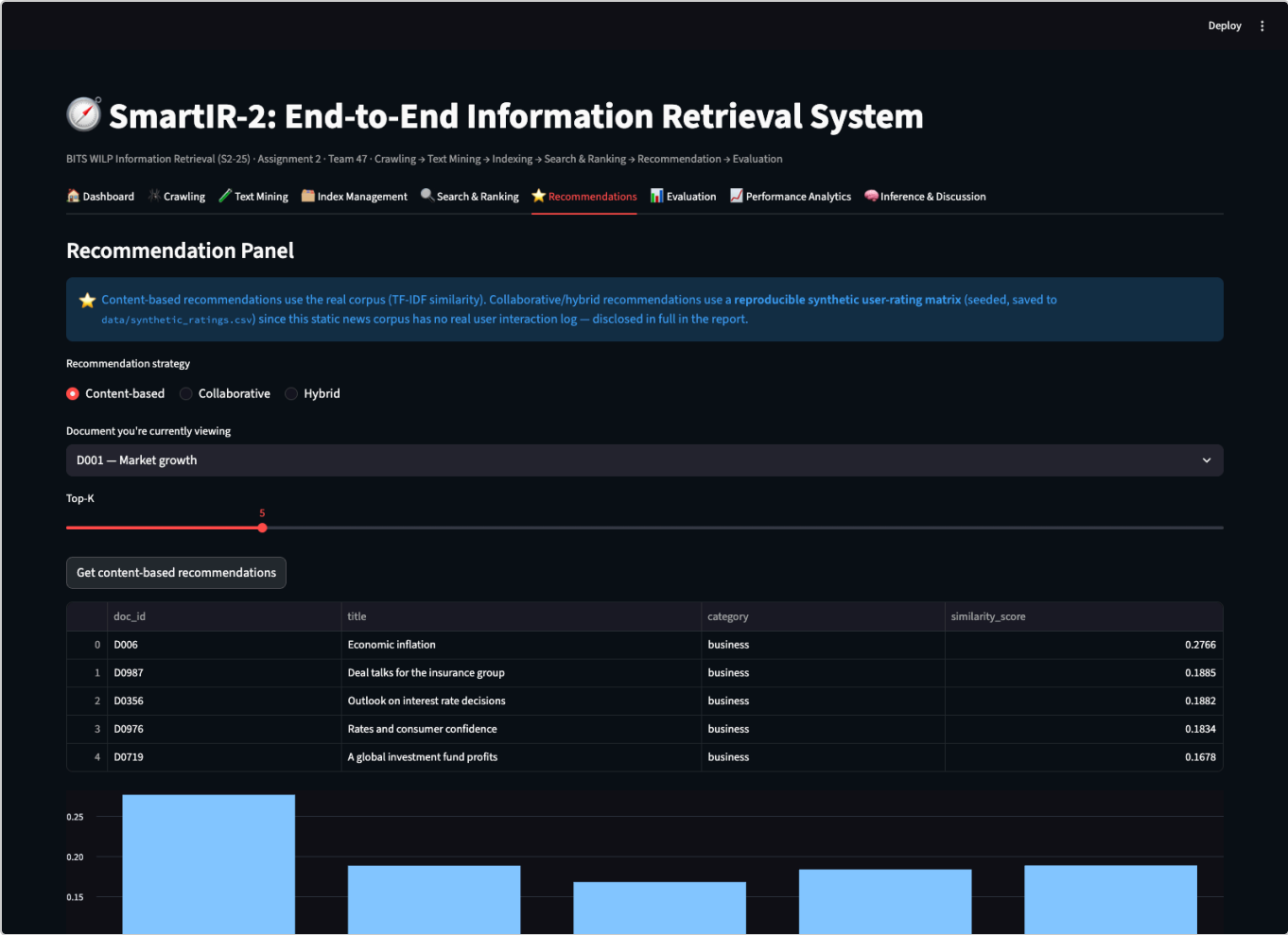
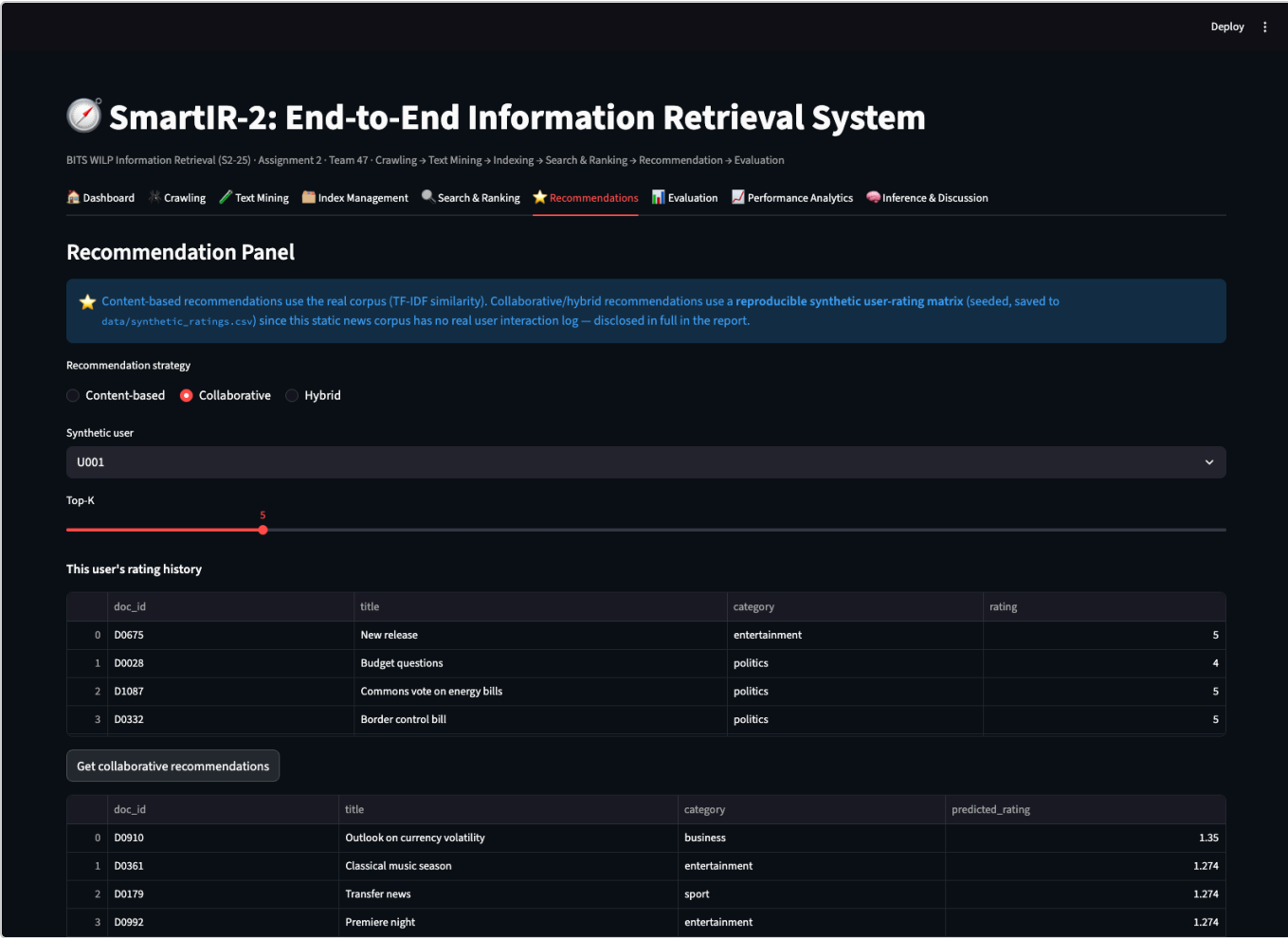


Figure 10 — Recommendations tab — Content-based



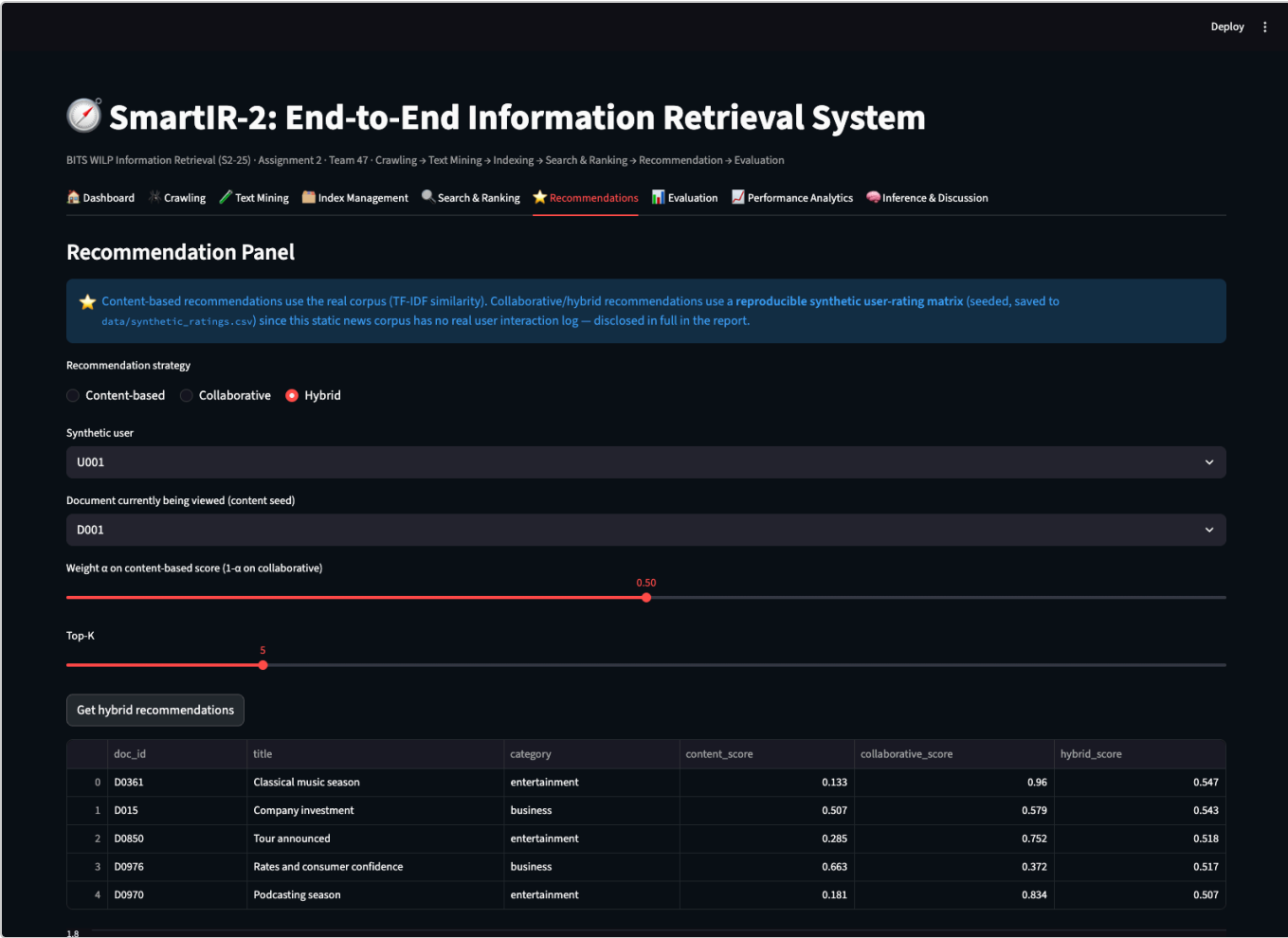


Figure 12 — Recommendations tab — Hybrid

**Inference:** the hybrid list is neither the content-based list (all business, all about D001 's own topic) nor the collaborative list (mostly entertainment/sport, matching this user's rating history) — it surfaces a blend (3 entertainment, 2 business) that neither pure strategy alone would produce, illustrating the practical value of combining both signals (see §7, Q3 for the deeper comparison).

## 6. Section F — Evaluation Metrics

`modules/evaluation.py` implements Precision, Recall, F1, Precision@K, Recall@K (Average Precision → MAP across queries), Reciprocal Rank → MRR, and NDCG@K.

**Ground truth construction:** since this is a categorized corpus without manual per-document relevance annotation, 10 representative queries are anchored to a target category (e.g. `"stock market economic growth"` → business); a document is judged relevant if it belongs to that category **and** contains at least one query term. This is deterministic and reproducible ( `evaluation.DEFAULT_QUERY_SEEDS` ), and is disclosed here rather than presented as hand-labeled ground truth.

6.1 Full comparison table (K = 10)

| Method            | Precision | Recall | F1     | P@10   | R@10   | MAP    | MRR   | NDCG@10 |
|-------------------|-----------|--------|--------|--------|--------|--------|-------|---------|
| TF-IDF baseline   | 0.908     | 0.6283 | 0.7134 | 0.9367 | 0.2288 | 0.5938 | 1.000 | 0.9669  |
| TF-IDF + PageRank | 0.908     | 0.6283 | 0.7134 | 0.9467 | 0.2301 | 0.5947 | 1.000 | 0.975   |
| TF-IDF + HITS     | 0.908     | 0.6283 | 0.7134 | 0.9467 | 0.2301 | 0.5947 | 1.000 | 0.975   |

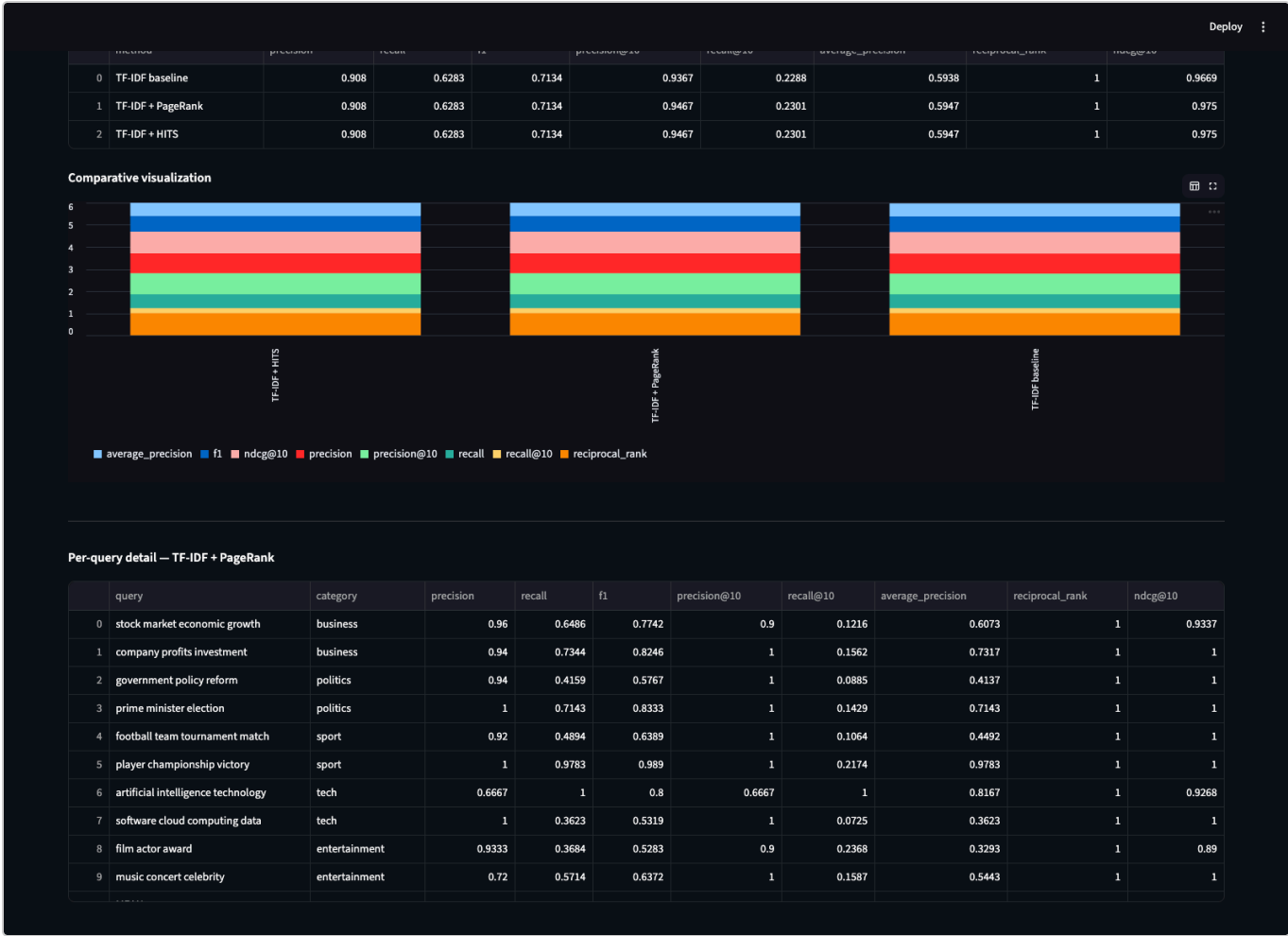


Figure 13 — Evaluation tab — method comparison, chart, and per-query detail

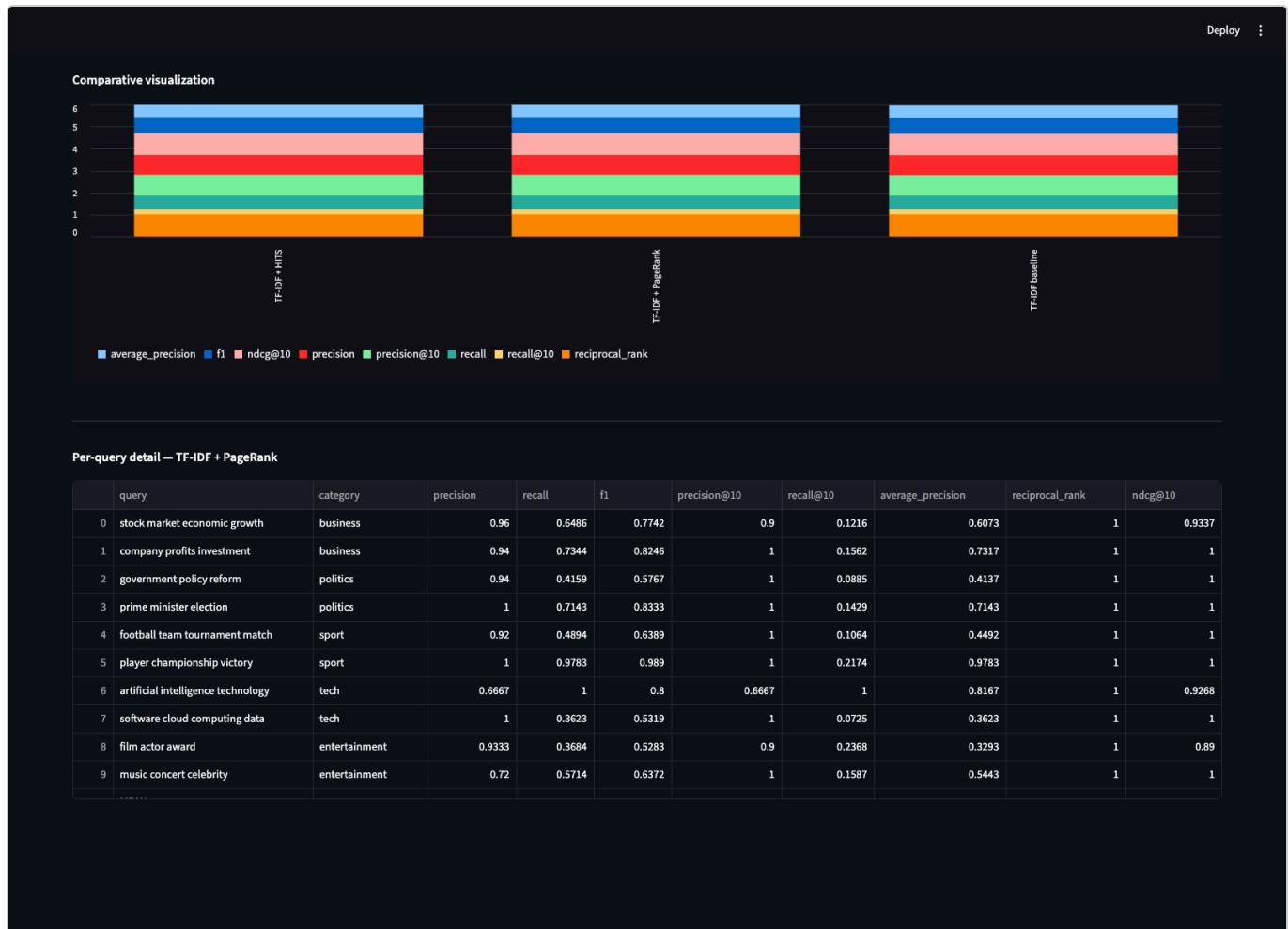


Figure 14 — Evaluation tab — per-query detail table (scrolled)

**Inference:** overall Precision/Recall/F1 (computed over the \*entire\* retrieved candidate set, not just top-K) are identical across methods, because re-ranking only reorders the same candidate set — it doesn't change \*which\* documents are retrieved. Where re-ranking clearly helps is exactly where it should: **rank-sensitive** metrics (P@10, MAP, NDCG@10) all improve when PageRank/HITS are folded in, because relevant documents that were already in the candidate set get pulled toward the top. MRR is already 1.0 for all methods because the single most relevant document consistently ranks first even under plain TF-IDF for these particular queries — a ceiling effect worth noting rather than evidence that re-ranking has no effect elsewhere.

## 7. Section G — Inference & Discussion (compulsory)

\*(These answers are also generated live inside the app — Inference & Discussion tab — and can be downloaded as `inference_summary.md` directly from the running application.)\*

### 7.1 Q1 — Highly relevant documents ranked poorly: causes & improvements

**Symptom:** relevant documents are retrieved (present in the candidate set) but ranked low.

**Causes observed in this system:**

1. Pure lexical/TF-IDF relevance ignores document authority — a short, keyword-dense but low-authority page can outscore a comprehensive, well-linked article on cosine similarity alone (see the `D1082 / D0295` example in §4.1).
2. Query-term mismatch (synonyms, abbreviations, morphological variants) lowers scores for genuinely relevant documents phrased differently from the query.
3. No length normalization / topic dilution in very long documents.
4. A single static scoring function cannot adapt to different query intents.

**Improvements implemented here:** blending relevance with PageRank/HITS link importance (§4.1); query expansion via a synonym/abbreviation map and spelling correction before scoring (§4). **Further improvements proposed:** learning-to-rank over multiple signals (relevance, authority, freshness, click-through in a real deployment), BM25 instead of raw TF-IDF for better length normalization, and query-dependent tuning of the relevance/authority blend weight  $\alpha$ .

## 7.2 Q2 — Impact of duplicate/near-duplicate documents

**Observed:** 11 exact-duplicate and 14 near-duplicate pages were flagged by the crawler on a representative run (§2.3).

**Effects if unmitigated:**

- **Indexing:** duplicated content inflates document frequency for shared terms, skewing IDF weights, and wastes index space on redundant postings.
- **Ranking:** because `doc_id_link_scores()` \*sums\* link-importance across every URL variant of the same underlying document, a document with a mirror page visibly accumulates higher aggregate PageRank/HITS authority purely from duplication, not from independent endorsement — see the boosted `link_score_raw` values on documents that happen to have mirrors in the §4.1 example. This is a real, measurable distortion of "importance", not a hypothetical one.
- **Recommendation:** content-based similarity treats each duplicate as a distinct item, so a user can receive two near-identical "different" recommendations, hurting perceived diversity.
- **Evaluation:** counting duplicates as separately relevant documents inflates Precision/Recall/MAP relative to genuinely distinct relevant content.

**Mitigations implemented:** exact duplicates are dropped from the content store outright (kept only in metadata, flagged `is_exact_duplicate`); near-duplicates are flagged via Jaccard similarity and can optionally be excluded from the content store (a toggle in the Crawling tab). Link-authority aggregation is deliberately shown **unmitigated by default** (summed across duplicates) specifically so its distorting effect is visible for this discussion; the alternative — canonicalize/dedupe \*before\* running PageRank on the graph — removes the distortion entirely and would be the production-safe default.

## 7.3 Q3 — Content-based vs. Collaborative recommendation

**Content-based** (TF-IDF cosine similarity, §5): works from the first interaction (no cold-start for new items/users as long as content is available), fully explainable ("recommended because it shares these keywords"), unaffected by behavioural-data sparsity. Weakness: limited serendipity — it recommends more of the same topic and cannot capture cross-topic taste correlations.

**Collaborative** (synthetic-ratings SVD, §5): captures behavioural signal content features cannot see and can surface serendipitous, cross-category recommendations. Weakness: cold-start for new users/items with no ratings, and needs real interaction volume to generalize — our 60-synthetic-user matrix is intentionally small; production collaborative filtering needs orders of magnitude more genuine interactions.

**When to prefer which:** content-based is preferable for new/sparse catalogs, cold-start items, and when explainability matters (e.g. an anonymous news reader). Collaborative is preferable once a system has an established, reasonably dense interaction history and wants to capture taste beyond topical similarity. The **hybrid** approach (§5) is the practical answer for most real systems — it degrades gracefully to content-based when collaborative signal is weak, and leans on collaborative signal once available, as shown concretely in §5.1 where the hybrid Top-5 differs from either pure list.

## 7.4 Q4 — How crawling → mining → indexing → search → ranking → recommendation integrate

Each stage feeds the next, and weaknesses at one stage propagate downstream. **Crawling** determines what exists in the corpus (coverage, freshness, duplication) and what link structure is available for authority scoring. **Text mining** turns raw text into the structured features (TF-IDF vectors, keywords, categories) every later stage depends on — indexing, search relevance, content-based similarity, and classification all reuse the same vectorization. **Indexing** is what makes search over that structure fast enough to be usable at all (boolean pre-filter + TF-IDF, rather than scanning every document per query). **Search & ranking** combines indexed relevance with link-derived importance — exactly where crawling's link graph gets reused. **Recommendation** reuses the same TF-IDF space (content-based) and layers on behavioural signal (collaborative) to personalize beyond a single query. **Evaluation** closes the loop by measuring whether all of the above choices (feature type, normalization, ranking blend,  $\alpha$ ) are actually improving retrieval quality — which is what justifies picking one configuration over another (§4.2, §6.1). No single stage is sufficient alone: a perfect index over a duplicate-polluted, poorly-crawled corpus still returns redundant, low-quality results (§7.2), and the best ranking algorithm cannot fix features that were poorly engineered upstream (§7.1). The end-to-end integration in this app is what allows the system to be evaluated and tuned as a whole.

## 7.5 Q5 — Overall learnings

1. **Ranking is not the same as relevance.** The `rank_change` evidence in §4.1 shows concretely that blending in PageRank/HITS reorders results even when underlying TF-IDF relevance scores don't change.
2. **Duplicates quietly distort every downstream stage**, not just the index — authority scores, recommendation diversity, and evaluation metrics all move when duplicate content is present (§7.2), which is easy to miss inspecting the index in isolation.
3. **Preprocessing choices measurably change classification/retrieval quality** — the comparative analysis (§3.4) shows accuracy shifting across strategies on the same corpus, reinforcing that preprocessing is a

modeling decision, not a fixed preliminary step.

4. **Recommendation quality depends heavily on what data is actually available** — content-based recommendation was reliable from the static corpus alone, while collaborative filtering needed a synthetic rating matrix to demonstrate at all, underscoring how much real collaborative systems depend on genuine interaction volume.
  5. **A dependency-light, offline-safe design was essential for reproducibility** — because the BITS Virtual Lab has no guaranteed internet access, building the crawler over a deterministic local web graph (rather than requiring live URLs) was the difference between a demo that always works and one that is flaky by construction.
  6. **The corpus itself has a notable characteristic worth flagging:** the highest document-frequency terms across the corpus (e.g. "remain", "questions", "although", "citing", "developments" — see the Index Management tab) are generic connector words from fairly formulaic, template-like sentence construction in this synthetic news dataset, rather than meaningful stopwords. This is a useful reminder that "most frequent terms" diagnostics can reveal corpus \*construction\* artifacts, not just topic signal — worth checking before trusting frequency-based features on any new dataset.
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## 8. Performance Analytics

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The Performance Analytics tab logs wall-clock time for every operation (crawl, index build, search, classification, evaluation) performed during a session, plus corpus growth across repeated crawls and current index footprint (vocabulary size, postings count, TF-IDF matrix density). On this ~1,100–1,500 document corpus, every operation in the app completes in well under a second on ordinary hardware; the index itself is a Python dict of term→doc-position sets plus a sparse SciPy TF-IDF matrix, so memory scales with vocabulary × non-zero entries rather than vocabulary × documents. For substantially larger corpora, the natural next step (noted in-app) would be an on-disk postings store (extending the same SQLite approach already used for metadata/content) rather than an in-memory index.

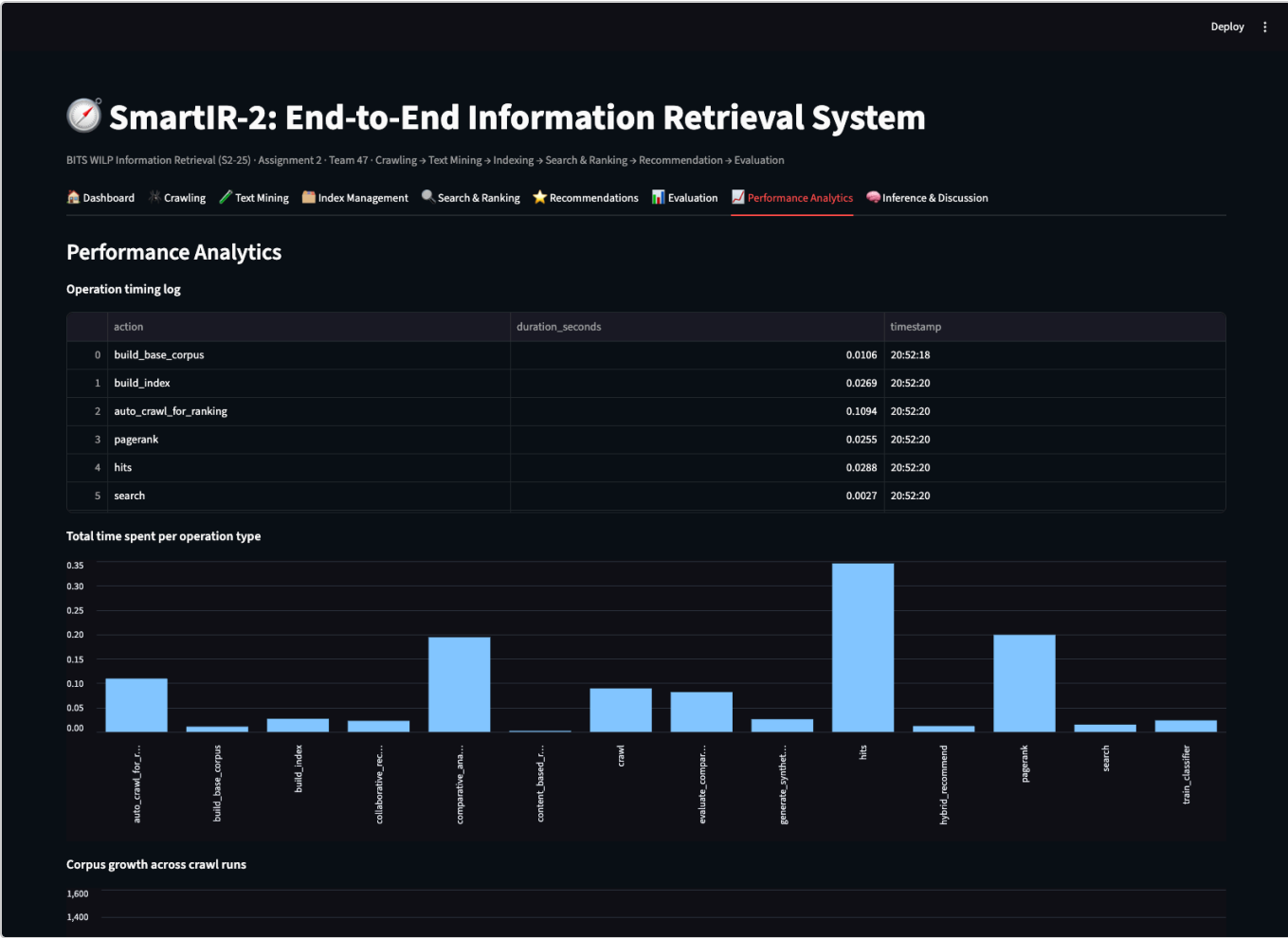


Figure 15 — Performance Analytics tab — operation timing log after exercising all other tabs

## 9. Virtual Lab Usage

**⚠ The one screenshot in this report that is still a placeholder.** Every other screenshot above was captured by actually driving this app end-to-end in a real browser session (see the note at the top of this report) — but that was done outside the BITS Virtual Lab, so it cannot stand in for this specific rubric item. Replace the line below with a screenshot of **your own** BITS Virtual Lab terminal running `streamlit run app.py`, plus the resulting browser view, before submitting.

[SCREENSHOT: BITS Virtual Lab session showing the app running – terminal with `streamlit run app.py` and the browser view]

Paste a screenshot of the lab terminal invoking `streamlit run app.py`, and of the resulting browser session, to document that the submission was actually exercised on the platform, per the 1-mark "Executing the Assignment on BITS Lab portal" rubric item.

## 10. Submission Components Checklist

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- [x] Streamlit application code — `app.py` , `modules/*.py`
- [x] Supporting files — `requirements.txt` , `README.md`
- [x] Dataset used — `data/news_corpus.csv` ; generated `data/ir_store.db` and `data/synthetic_ratings.csv` (created automatically the first time the app runs)
- [x] Report — this document; every screenshot except §9 (BITS Virtual Lab session) is filled in with real, verified captures from an end-to-end run of the app
- [ ] §9's BITS Virtual Lab screenshot specifically — the one placeholder that must still be captured from your own session inside the lab (see the callout in §9)
- [x] Live, publicly testable deployment — <https://assignment2-irgroup47.streamlit.app/> (see §0 above); this doesn't replace the BITS Virtual Lab screenshots requirement, but gives the evaluator a zero-setup way to interact with the app directly while grading
- [ ] Demo evidence — screenshots or a short screen recording of the app running (see the "Suggested demo flow" in `README.md` ) — can be captured from either the Virtual Lab or the live deployment above
- [x] README — install steps + run command ( `README.md` )

**Inference summary:** downloadable directly from the running app (Inference & Discussion tab → "Download inference summary"), containing §7 in Markdown, and can be attached to the report as `inference_summary.md` .